

● HARD

● PROBLEM-SOLVING AND DATA ANALYSIS

Percentages

10 questions · ~15 min

02

Q1

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS

A gift shop buys souvenirs at a wholesale price of 7.00 dollars each and resells them each at a retail price that is 290% of the wholesale price. At the end of the season, any remaining souvenirs are marked at a discounted price that is 80% off the retail price. What is the discounted price of each remaining souvenir, in dollars?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q2

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS

210 is $p\%$ greater than 30. What is the value of p ?

STUDENT-PRODUCED RESPONSE

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$\cdot$ $\div$ $\cdot$ $\div$ $\cdot$

0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q3

PROBLEM-SOLVING AND DATA ANALYSIS

The positive number a is 2,241% of the sum of the positive numbers b and c , and b is 83% of c . What percent of b is a ?

- A. 23.24%
- B. 49.41%
- C. 2,324%
- D. 4,941%

Q4

PROBLEM-SOLVING AND DATA ANALYSIS

The result of increasing the quantity x by 400% is 60. What is the value of x ?

- A. 12
- B. 15
- C. 240
- D. 340

Q5

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS

The number of zebras in a population in 2018 was 1.27 times the number of zebras in this population in 2014. If the number of zebras in this population in 2014 is $p\%$ of the number of zebras in this population in 2018, what is the value of p , to the nearest whole number?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q6

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS

The number a is 55% less than the number b . The number b is 320% greater than 160. What is the value of a ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q7

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS

A reseller buys certain books for a purchase price of 5.00 dollars each and then marks them each for sale at a consumer price that is 270% of the purchase price. After 4 months, any remaining books not yet sold are marked at a discounted price that is 70% off the consumer price. What is the discounted price of each of the remaining books, in dollars?

STUDENT-PRODUCED RESPONSE

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0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q8

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS

A school district is forming a committee to discuss plans for the construction of a new high school. Of those invited to join the committee, 15% are parents of students, 45% are teachers from the current high school, 25% are school and district administrators, and the remaining 6 individuals are students. How many more teachers were invited to join the committee than school and district administrators?

STUDENT-PRODUCED RESPONSE

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0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q9

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS

The number a is 70% less than the positive number b . The number c is 80% greater than a . The number c is how many times b ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q10

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS

The weight of sample A is 468% of the weight of sample B, and the weight of sample A is 0.780% of the weight of sample C. If the weight of sample C is $p\%$ of the weight of sample B, what is the value of p ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.